

PROCESS & SESSION

➤ How to check total/active/inactive session?

In Oracle Database DBA, you can check all active & inactive session using the following commands:

Total Session

```
SQL> SELECT COUNT(*) AS TOTAL_SESSION FROM V$SESSION;  
  
TOTAL_SESSION  
-----  
83
```

Active Session

```
SQL> SELECT COUNT(*) AS ACTIVE_SEEION FROM V$SESSION WHERE STATUS='ACTIVE';  
  
ACTIVE_SEEION  
-----  
83
```

Inactive Session

```
SQL> SELECT COUNT(*) AS INACTIVE_SESSION FROM V$SESSION WHERE STATUS='INACTIVE';  
  
INACTIVE_SESSION  
-----  
0
```

ALL Session

```
SQL> SELECT COUNT(*) AS total_sessions,  
2 SUM(CASE WHEN status = 'ACTIVE' THEN 1 ELSE 0 END) AS active_sessions,  
3 SUM(CASE WHEN status = 'INACTIVE' THEN 1 ELSE 0 END) AS inactive_sessions  
4 FROM v$session;  
  
TOTAL_SESSIONS ACTIVE_SESSIONS INACTIVE_SESSIONS  
-----  
82 82 0
```

➤ How to identify and kill inactive session?

STEP 1: USER 1 UPDATED THE JOB COLUMN WHO IS CALLED MILLER.

```
SQL> select * from emp;
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	SMITH	CLERK	7902	17-DEC-80	800		20
7499	ALLEN	SALESMAN	7698	20-FEB-81	1600	300	30
7521	WARD	SALESMAN	7698	22-FEB-81	1250	500	30
7566	JONES	MANAGER	7839	02-APR-81	2975		20
7654	MARTIN	SALESMAN	7698	28-SEP-81	1250	1400	30
7698	BLAKE	MANAGER	7839	01-MAY-81	2850		30
7782	CLARK	MANAGER	7839	09-JUN-81	2450		10
7788	SCOTT	ANALYST	7566	19-APR-87	3000		20
7839	KING	PRESIDENT		17-NOV-81	5000		10
7844	TURNER	SALESMAN	7698	08-SEP-81	1500	0	30
7876	ADAMS	CLERK	7788	23-MAY-87	1100		20
7900	JAMES	CLERK	7698	03-DEC-81	950		30
7902	FORD	ANALYST	7566	03-DEC-81	3000		20
7934	MILLER	CLERK	2134	23-JAN-82	1300		10

```
14 rows selected.  
  
SQL> update emp set JOB='IT' where ENAME='MILLER';  
  
1 row updated.
```

STEP 2: AT THE SAME TIME USER 2 UPDATED THE SAME ROW BUT SESSION IS BLOCKED.

```
SQL> select * from emp;
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	SMITH	CLERK	7902	17-DEC-80	800		20
7499	ALLEN	SALESMAN	7698	20-FEB-81	1600	300	30
7521	WARD	SALESMAN	7698	22-FEB-81	1250	500	30
7566	JONES	MANAGER	7839	02-APR-81	2975		20
7654	MARTIN	SALESMAN	7698	28-SEP-81	1250	1400	30
7698	BLAKE	MANAGER	7839	01-MAY-81	2850		30
7782	CLARK	MANAGER	7839	09-JUN-81	2450		10
7788	SCOTT	ANALYST	7566	19-APR-87	3000		20
7839	KING	PRESIDENT		17-NOV-81	5000		10
7844	TURNER	SALESMAN	7698	08-SEP-81	1500	0	30
7876	ADAMS	CLERK	7788	23-MAY-87	1100		20
7900	JAMES	CLERK	7698	03-DEC-81	950		30
7902	FORD	ANALYST	7566	03-DEC-81	3000		20
7934	MILLER	CLERK	2134	23-JAN-82	1300		10

```
14 rows selected.  
  
SQL> update emp set JOB='IT ANALYST' where ENAME='MILLER';  
█
```

STEP 3: NOW IDENTIFY THE BLOCKING SESSION

```
SQL> SELECT SID,SERIAL#,USERNAME,STATUS FROM V$SESSION WHERE STATUS='INACTIVE' AND TYPE='USER';
```

SID	SERIAL#	USERNAME	STATUS
20	40901	SCOTT	INACTIVE

STEP 4: RELEASE THIS BLOCKING SESSION

```
SQL> ALTER SYSTEM KILL SESSION '20,40901';  
System altered.
```

STEP 5: USER IS NOW RELEASED THE BLOCK.

```
SQL> SELECT SID,SERIAL#,USERNAME,STATUS FROM V$SESSION WHERE STATUS='INACTIVE' AND TYPE='USER';  
no rows selected
```

```
SQL> update emp set JOB='IT ANALYST' where ENAME='MILLER';  
update emp set JOB='IT ANALYST' where ENAME='MILLER'  
*  
ERROR at line 1:  
ORA-12899: value too large for column "SCOTT"."EMP"."JOB" (actual: 10, maximum: 9)
```

ALTER SYSTEM KILL SESSION → Marks session for cleanup by Oracle.

kill -9 → Forcefully kills Oracle OS process from Linux. Use only if Oracle session is stuck

➤ **How to kill multiple inactive session via script?**

STEP 1: create the shell script

```
[oracle@localhost u01]$  
[oracle@localhost u01]$ cat kill_inactive.sh  
#!/bin/bash  
  
export ORACLE_SID=orcl  
. oraenv <<< $ORACLE_SID  
  
sqlplus -s / as sysdba <<EOF  
  
SET SERVEROUTPUT ON  
SET HEADING OFF  
SET FEEDBACK OFF  
  
BEGIN  
  FOR r IN (  
    SELECT sid, serial#  
    FROM v$session  
    WHERE status='INACTIVE'  
    AND username IS NOT NULL  
    AND last_call_et > 600  
  )  
  LOOP  
    EXECUTE IMMEDIATE  
      'ALTER SYSTEM KILL SESSION '''||  
      r.sid||','||r.serial#||''' IMMEDIATE';  
  
    DBMS_OUTPUT.PUT_LINE('Killed Session: '''||  
                          r.sid||','||r.serial#);  
  END LOOP;  
END;  
/  
  
EXIT;  
EOF  
[oracle@localhost u01]$
```

STEP 2: User 1 update the MGR column who is KING

```
SQL> select * from emp;
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	SMITH	CLERK	7902	17-DEC-80	800		20
7499	ALLEN	SALESMAN	7698	20-FEB-81	1600	300	30
7521	WARD	SALESMAN	7698	22-FEB-81	1250	500	30
7566	JONES	MANAGER	7839	02-APR-81	2975		20
7654	MARTIN	SALESMAN	7698	28-SEP-81	1250	1400	30
7698	BLAKE	MANAGER	7839	01-MAY-81	2850		30
7782	CLARK	MANAGER	7839	09-JUN-81	2450		10
7788	SCOTT	ANALYST	7566	19-APR-87	3000		20
7839	KING	PRESIDENT		17-NOV-81	5000		10
7844	TURNER	SALESMAN	7698	08-SEP-81	1500	0	30
7876	ADAMS	CLERK	7788	23-MAY-87	1100		20
7900	JAMES	CLERK	7698	03-DEC-81	950		30
7902	FORD	ANALYST	7566	03-DEC-81	3000		20
7934	MILLER	CLERK	2134	23-JAN-82	1300		10

14 rows selected.

```
SQL> update emp set MGR=7645 where ENAME='KING';
```

1 row updated.

STEP 3: At the same time USER 2 update the same row but the session is blocked.

```
SQL> select * from emp;
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	SMITH	CLERK	7902	17-DEC-80	800		20
7499	ALLEN	SALESMAN	7698	20-FEB-81	1600	300	30
7521	WARD	SALESMAN	7698	22-FEB-81	1250	500	30
7566	JONES	MANAGER	7839	02-APR-81	2975		20
7654	MARTIN	SALESMAN	7698	28-SEP-81	1250	1400	30
7698	BLAKE	MANAGER	7839	01-MAY-81	2850		30
7782	CLARK	MANAGER	7839	09-JUN-81	2450		10
7788	SCOTT	ANALYST	7566	19-APR-87	3000		20
7839	KING	PRESIDENT		17-NOV-81	5000		10
7844	TURNER	SALESMAN	7698	08-SEP-81	1500	0	30
7876	ADAMS	CLERK	7788	23-MAY-87	1100		20
7900	JAMES	CLERK	7698	03-DEC-81	950		30
7902	FORD	ANALYST	7566	03-DEC-81	3000		20
7934	MILLER	CLERK	2134	23-JAN-82	1300		10

14 rows selected.

```
SQL> update emp set COMM=1200 where ENAME='KING';
```

STEP 4: Now run shell script the session is released

```
[oracle@localhost u01]$  
[oracle@localhost u01]$ ./kill_inactive.sh  
ORACLE_SID = [orcl] ? The Oracle base remains unchanged with value /u01/app/oracle  
Killed Session: 20,8775  
Killed Session: 390,41814
```

➤ Commands to check blocking session?

STEP 1: User 1 update the first row on the emp table.

```
SQL> select * from emp;
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	SMITH	CLERK	7902	17-DEC-80	800		20
7499	ALLEN	SALESMAN	7698	20-FEB-81	1600	300	30
7521	WARD	SALESMAN	7698	22-FEB-81	1250	500	30
7566	JONES	MANAGER	7839	02-APR-81	2975		20
7654	MARTIN	SALESMAN	7698	28-SEP-81	1250	1400	30
7698	BLAKE	MANAGER	7839	01-MAY-81	2850		30
7782	CLARK	MANAGER	7839	09-JUN-81	2450		10
7788	SCOTT	ANALYST	7566	19-APR-87	3000		20
7839	KING	PRESIDENT		17-NOV-81	5000		10
7844	TURNER	SALESMAN	7698	08-SEP-81	1500	0	30
7876	ADAMS	CLERK	7788	23-MAY-87	1100		20
7900	JAMES	CLERK	7698	03-DEC-81	950		30
7902	FORD	ANALYST	7566	03-DEC-81	3000		20
7934	MILLER	CLERK	2134	23-JAN-82	1300		10

14 rows selected.

```
SQL> update emp set SAL=9000 where empno=7369;
```

1 row updated.

STEP 2: At the same time USER 2 update the same row but the session is blocked.

```
SQL> select * from emp;
```

EMPNO	ENAME	JOB	MGR	HIREDATE	SAL	COMM	DEPTNO
7369	SMITH	CLERK	7902	17-DEC-80	800		20
7499	ALLEN	SALESMAN	7698	20-FEB-81	1600	300	30
7521	WARD	SALESMAN	7698	22-FEB-81	1250	500	30
7566	JONES	MANAGER	7839	02-APR-81	2975		20
7654	MARTIN	SALESMAN	7698	28-SEP-81	1250	1400	30
7698	BLAKE	MANAGER	7839	01-MAY-81	2850		30
7782	CLARK	MANAGER	7839	09-JUN-81	2450		10
7788	SCOTT	ANALYST	7566	19-APR-87	3000		20
7839	KING	PRESIDENT		17-NOV-81	5000		10
7844	TURNER	SALESMAN	7698	08-SEP-81	1500	0	30
7876	ADAMS	CLERK	7788	23-MAY-87	1100		20
7900	JAMES	CLERK	7698	03-DEC-81	950		30
7902	FORD	ANALYST	7566	03-DEC-81	3000		20
7934	MILLER	CLERK	2134	23-JAN-82	1300		10

14 rows selected.

```
SQL> update emp set SAL=8000 where empno=7369;
```

█

STEP 3: Check the blocking session

```
SQL> show user
USER is "SYS"
SQL>
SQL>
SQL> SELECT blocking_session AS BLOCKER_SID, SID AS BLOCKED_SID, SERIAL#, WAIT_CLASS, SECONDS_IN_WAIT FROM V$SESSION WHERE BLOCKING_SESSION IS NOT NULL;
```

BLOCKER_SID	BLOCKED_SID	SERIAL#	WAIT_CLASS	SECONDS_IN_WAIT
390	471	16338	Application	297

STEP 4: Released the blocking session

```
SQL> ALTER SYSTEM KILL SESSION '471,16338';  
System altered.
```

```
SQL> update emp set SAL=8000 where empno=7369;  
update emp set SAL=8000 where empno=7369  
*  
ERROR at line 1:  
ORA-00028: your session has been killed
```

➤ How to increase process parameters?

Check the process

```
SQL> SELECT NAME,VALUE FROM V$PARAMETER WHERE NAME='processes';  
  
NAME                                     VALUE  
-----  
processes                               300  
  
SQL> SHOW PARAMETER PROCESS;  
  
NAME                                TYPE        VALUE  
-----  
aq_tm_processes                    integer      1  
cell_offload_processing             boolean      TRUE  
db_writer_processes                integer      1  
gcs_server_processes               integer      0  
global_txn_processes               integer      1  
job_queue_processes                integer      20  
log_archive_max_processes           integer      4  
processes                          integer      300  
processor_group_name                string
```

Change the process value.

```
SQL> ALTER SYSTEM SET processes=200 SCOPE=SPFILE;  
System altered.  
  
SQL> SHU IMMEDIATE  
Database closed.  
Database dismounted.  
ORACLE instance shut down.  
SQL>  
SQL> STARTUP  
ORACLE instance started.  
  
Total System Global Area  520091504 bytes  
Fixed Size                 8898416 bytes  
Variable Size              243269632 bytes  
Database Buffers           264241152 bytes  
Redo Buffers                3682304 bytes  
Database mounted.  
Database opened.  
SQL>  
SQL>  
SQL> SELECT NAME,VALUE FROM V$PARAMETER WHERE NAME='processes';  
  
NAME                                     VALUE  
-----  
processes                               200
```